

PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

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Abstract

method across tasks.

038

001 Segmentation models in medical imaging are trained on one
002 acquisition setting and must generalize to others. Label-
003 generative domain randomization, of which SynthSeg is the
004 archetype, addresses this by discarding the acquired scan
005 and resampling intensities from a per-label Gaussian mixture
006 over a dense anatomical label map – an annotation far
007 richer than the segmentation task itself provides, and one
008 that does not preserve texture: real gradients and partial-
009 volume detail become piecewise noise. We ask when that
010 loss costs accuracy, and find the answer is a property of
011 the task. Where the target is separated from its surround-
012 ings by an anatomical interface – encapsulated abdominal
013 organs, bounded by fat planes – destroying texture costs
014 little. Where no interface exists – diffuse glioma, infil-
015 trating beyond both the enhancing margin and the peri-
016 tumoral FLAIR abnormality – the target must be inferred
017 from tissue appearance, and destroying texture can be ex-
018 pensive. We exploit this with PALETTE, a single-source
019 augmentation needing no dense label map: it partitions
020 a scan’s foreground by unsupervised k-means, sub-divides
021 each class spatially by a Voronoi partition, and applies a
022 signed random affine remap to the real intensities of each
023 sub-region. Remapping existing intensities rather than re-
024 placing them leaves edges, gradients and partial-volume
025 detail largely intact; only the contrast relationships be-
026 tween regions are randomized. A controlled ablation iso-
027 lates this: holding the partition and per-region statistics
028 fixed and swapping only the real-intensity remap for label-
029 style noise fill, across fourteen training settings, every large
030 effect falls on a target without a tissue interface (+13.3
031 and +7.2 Dice on brain tumour, +7.7 on MS lesions),
032 while no interface-bounded target gains more than 1.4;
033 not every appearance-defined target shows a gain. Across
034 cross-contrast, cross-modality and cross-dataset transfer on
035 eight tasks and sixteen datasets, one nnU-Net configuration
036 trained with PALETTE is the best method overall: best Dice
037 on six tasks and significantly better than every competing

1. Introduction

039

040 Medical image segmentation models are trained on one ac-
041 quisition setting and are frequently applied to another: a
042 different scanner, sequence, or imaging modality than the
043 one seen during training. MRI contrast is a protocol choice
044 rather than a physical constant, so the same anatomy yields
045 incompatible intensity distributions across sequences, and a
046 network that has only seen T1-weighted brain images can
047 fail outright on T2-weighted or FLAIR images of the same
048 anatomy [3, 34]. Acquiring and annotating training data
049 for every contrast a model might encounter is not practical:
050 manual segmentation is expensive, and clinical sites rarely
051 share raw data across institutions [33, 52]. The single-
052 source domain generalization setting — train on one la-
053 belled dataset in one contrast, generalize to unseen contrasts
054 and modalities without any target-domain data — is there-
055 fore the setting that matters for deployment, and the one a
056 growing body of work addresses [36, 46, 56, 58].

057 The dominant single-source strategy is label-generative
058 domain randomization, popularized by SynthSeg [3]. Given
059 a dense anatomical label map — a voxel-wise parcellation
060 of the whole image, not the sparse structure-of-interest an-
061 notation a segmentation dataset normally provides — in-
062 tensities are resampled from a per-label Gaussian mixture,
063 producing images of arbitrary, unrealistic contrast that force
064 the network to ignore contrast and rely on shape and lay-
065 out [3]. This is effective, but carries two structural costs.
066 First, it is *label-driven*: the dense parcellation it resamples
067 from is a richer annotation than the task itself requires, so
068 generating synthetic contrast costs exactly the expensive la-
069 labelling the method aims to reduce reliance on. Second, and
070 less discussed, it *does not preserve texture*: replacing every
071 labelled region with piecewise Gaussian noise discards the
072 real gradients, partial-volume boundaries, and unlabelled
073 fine structure the acquired scan contains. Layout survives;
074 texture does not.

We ask whether discarding texture is necessary for contrast robustness, or merely the path of least resistance once a dense label map is assumed. Our central finding is that the answer depends on the *task*: large gains from preserving real intensity structure occur only where the segmentation target is defined by tissue appearance rather than separated from its surroundings by a genuine anatomical interface. That texture carries diagnostic signal in exactly the tasks where no such interface exists is established independently of us — texture features discriminate MS lesions from normal-appearing white matter [57], and diffuse glioma infiltrates beyond every margin conventional MRI shows [17, 48] — and our contribution is to show that this distinction governs where a texture-preserving augmentation can help substantially.

We exploit it with **PALETTE**, a single-source augmentation that needs no dense label map. **PALETTE** partitions the foreground of a scan into intensity classes with unsupervised 1-D k -means, further subdivides each class into spatial sub-regions with a Voronoi partition, and applies a signed random affine remap to the *real* intensities within each sub-region. Because the transform remaps existing intensities rather than replacing them with sampled noise, edges, gradients, and partial-volume detail largely survive within each sub-region; only the region-to-region contrast relationship is randomized, including sign inversions no physical acquisition protocol produces. The optional final step is the only one that uses labels, and it uses only the sparse task labels already present in any segmentation dataset — never a dense parcellation.

We test the mechanism claim directly rather than inferring it from aggregate scores. Holding the k -means/Voronoi partition and the per-region statistics fixed, we replace only the real-intensity affine remap with label-style noise fill, matching SynthSeg’s resampling inside **PALETTE**’s own geometry, across fourteen training settings. Every large effect of that swap falls on a target without a tissue interface — +13.3 and +7.2 Dice on brain tumour, +7.7 on MS lesions — while no interface-bounded target gains more than 1.4. The converse does not hold: several appearance-defined settings, breast lesions among them, show no reliable effect, so we state the result in one direction only (§4.3). An independent, segmentation-free gradient-orientation measurement confirms that texture is what changes at the fill-swap step: the noise-fill steps leave it at the noise floor while Dice still rises, and only the real-fill swap restores it.

Empirically, training a single nnU-Net [22] configuration — fixed across datasets, with only the epoch budget varying — we evaluate cross-contrast, cross-modality, and cross-dataset transfer on eight segmentation tasks: healthy brain, MS lesion, brain tumour, abdominal organ, mandible, breast lesion, spine and pelvis. We com-

pare against a no-augmentation baseline, both variants of SynthSeg (the original dense-label-driven variant, and an EM variant with no public implementation that we re-implement), SRCSM [49], and Auglab [34], a published augmentation pipeline designed for cross-modality training and the strongest competitor. **PALETTE** is the best method overall: it attains the best Dice on six of the eight tasks and is significantly better than every competitor across tasks, on both Dice and boundary distance. No competitor is reliably second: the strongest alternative switches between Auglab and SRCSM from task to task, so **PALETTE** is the only method at or near the top throughout.

In summary, we make the following contributions:

- **A task-level account of when texture preservation matters.** Large gains from preserving real intensity structure occur only where the segmentation target has no anatomical interface separating it from its surroundings. We test this with a controlled intervention rather than asserting it post hoc, and report where it does not extend.
- **A causal ablation isolating texture.** Holding the intensity and spatial partition and the per-region statistics fixed and swapping only the real-intensity remap for label-style noise fill, across fourteen training settings, confines every large effect to no-interface targets. An independent gradient-orientation measurement confirms that texture, not the partitioning scheme, is what changes at that step.
- **PALETTE**, the single-source contrast augmentation that exploits this, requiring no dense anatomical label map and using only the sparse task labels a segmentation dataset already provides.
- **Breadth evidence.** Across cross-contrast, cross-modality and cross-dataset transfer on eight tasks and sixteen datasets, with one configuration reused throughout, **PALETTE** is best on six tasks and significantly better than every competitor across tasks. Where it does not win we report whether the method ahead is actually ahead: on healthy brain it is a tie, on mandible a real loss.

2. Related Work

2.1. Domain generalization for medical segmentation

Domain generalization (DG) targets unseen distributions without access to target data, unlike unsupervised domain adaptation, which assumes unlabelled target images and solves an easier problem [45]. Most medical DG methods are *multi-source*, interpolating across several labelled domains by meta-learning, feature alignment, frequency-space exchange, multi-frequency architectures, foundation-model scale, or diffusion-based stylization [2, 30, 33, 35, 43, 52] — all of which still rely on source diversity. We instead work in the *single-source* regime — one labelled dataset, no target data, no second source — the setting that

matters when target-domain data is unavailable at training time, which is routine in clinical deployment [36, 58]. This rules out cross-domain interpolation and makes architecture-coupled methods orthogonal to our contribution: our mechanism acts at the input level and leaves the segmentation network a fixed nnU-Net [22].

2.2. Synthesis-based contrast-agnostic training

The dominant single-source strategy is label-generative domain randomization, introduced by SynthSeg [3]: intensities are sampled from a per-label Gaussian mixture over an anatomical segmentation, yielding images of arbitrary contrast that force contrast-invariance. This family carries two structural costs: it is *label-driven* (a dense anatomical label map is required per subject – SIAM shifts the dependence from label *quantity* to *quality* with curated templates, but each still needs a full dense parcellation [50]), and it is *texture-destroying*, since per-label Gaussian sampling replaces real gradients, partial-volume detail and unlabelled structure with piecewise noise, retaining layout but not texture. The lineage has been specialized to white-matter and MS lesions and to atlas-based expectation-maximization [6, 7, 28]. Among SynthSeg’s own variants we take the original (*noEM*), which resamples intensities from a dense anatomical label map and never looks at the acquired scan, and the *EM* variant, which instead fits its per-class intensity model to the real image by expectation-maximization and so needs only the sparse task labels a segmentation dataset already provides, not a dense parcellation. The EM variant has no released implementation; ours follows the mechanism described in the original paper [3], and a faithful re-implementation is itself a contribution.

2.3. Texture as a segmentation signal

That local texture, not only intensity level, carries information a segmentation model can use is established independently of us: second-order gray-level statistics discriminate image content beyond simple intensity [18], and quantitative texture features from tumour regions carry prognostic signal across CT, MRI and PET [27]. Closer to our setting, Kang et al. disentangle structure from texture in cross-domain image translation with a texture co-occurrence loss, targeting the same failure mode we do — naive translation distorts fine texture and degrades downstream segmentation [24]; a texture-fidelity loss has likewise been added to GAN-based lesion augmentation [15]. That some targets resist boundary-based methods for lack of a sharp edge, and are better served by texture-based segmentation, has been argued directly for diffuse white-matter lesions [26]. None of this prior work, to our knowledge, connects texture preservation in an *augmentation* to *which* segmentation targets it helps; that connection, tested causally, is our contribution.

2.4. Single-source intensity and appearance augmentation

Where the first line of single-source methods synthesizes from a label map, a second manufactures appearance diversity directly from the real image, with no generative label model. General-purpose libraries stack intensity, spatial and artifact transforms on an nnU-Net trainer; the recent Auglab pipeline [22, 34] is one such GPU-accelerated framework and is the strongest competitor in our experiments, and Bi-gAug stacks a comparable set of coarse intensity, quality and spatial perturbations under the same single-source DG protocol we use [58]. Our method inserts the PALETTE transform as one additional operator inside the Auglab pipeline, so the comparison against Auglab directly isolates PALETTE’s contribution. Closer to our operator are random-filter methods, which vary appearance with an untrained nonlinear filter: GIN applies randomly re-initialized convolutions for label-free appearance variants [36] (the GIN-alone configuration we compare against is an ablation in the original paper of a two-view objective on 2-D slices, so our comparison is illustrative of the random-filter family rather than a reproduction of that paper’s proposed method); related variants establish the random-filter principle for natural images, perturb intensities per class using the task mask, add learned diffusion generators, or perturb the Fourier amplitude spectrum instead of the image itself [8, 31, 40, 46, 56, 59]. Our operator instead partitions the foreground into intensity classes and Voronoi sub-regions and applies a signed *affine* remap of the real intensities within each, invertible and gradient-preserving by construction. The closest peer is SRCSM [49] (semantic-aware random convolution with source matching), which also localizes its perturbation region-wise in true 3-D and, like PALETTE, layers exactly one label-consuming component over an otherwise label-free step: at training time it draws a fresh untrained convolution per anatomical class from the ground-truth map (needing a Gaussian smoothing pass to control the boundary-contrast artifacts this introduces), and at test time applies a label-free histogram matching onto the source distribution. The difference from PALETTE is not label use versus none, but *how* that step transforms the signal: an untrained nonlinear convolution versus a deterministic affine remap that leaves relative intensity structure inside each region intact.

2.5. Cross-contrast evaluation protocol

We evaluate a model trained on a single real input modality on every held-out modality of the *same* segmentation task, including the training modality itself, across eight tasks spanning brain, abdominal, dental, breast, spine and pelvic anatomy in MRI, CT and CBCT. This single-sequence-in, every-sequence-out design is not unprecedented – Auglab’s own evaluation follows the identical protocol for spine seg-

mentation [34] – and our contribution is breadth (eight tasks, not one) and a further axis: wherever an external cohort exists, it is pooled into its task as additional held-out contrasts from other sites, so the training dataset’s own test set is one point among several rather than the only held-out evidence. For mandible, which is acquired as CBCT only, that cross-dataset transfer is the entire held-out axis.

3. Method

We describe PALETTE (§3.1), how it is applied during training (§3.2), and the causal ablation used to isolate texture preservation as the source of its effect (§3.3).

3.1. The PALETTE transform

Let x denote the intensity volume of a training scan, min-max normalised to $[0, 1]$, and $F \subset \Omega$ its foreground voxels (an intensity threshold; $x_v > 0.01$ in our implementation). PALETTE applies four steps at every training iteration, illustrated end-to-end in Fig. 1 on one real CHAOS (abdominal MRI) and one real OnHarmony (brain MRI) volume. The figure’s five panels (a)–(e) are deliberately lettered rather than numbered 1–5 so they are never confused with the four method steps below: panels are *intermediate diagnostic renderings*, one per step plus the input, not a one-to-one visual walkthrough of the steps themselves (panel (e), the final training input, is the only panel that shows the texture the transform actually produces; (b)–(d) show flat preview values, defined in the caption).

1. Intensity partition. We run 1-D k -means on the foreground intensities $\{x_v : v \in F\}$ with k sampled per iteration from $\{2, \dots, 6\}$, producing k intensity classes $C_1, \dots, C_k$ (Fig. 1, panel (b)). This partition is unsupervised and requires no anatomical label map: it groups voxels by intensity alone, so the same operator applies unmodified to any input image regardless of its modality or contrast — CT, T1, T2, FLAIR, or otherwise. This step is occasionally skipped in favour of a single global remap (step 3 below, with $C = 1$); per-step probabilities are set once and never tuned per dataset, and are listed in Sec. 7.

2. Spatial sub-parcellation. Each intensity class C_i is further subdivided into spatial sub-regions by a Voronoi partition seeded with points sampled within C_i (Fig. 1, panel (c) shows the combined effect of this step and the remap of step 3), giving sub-regions $R_{i,1}, \dots, R_{i,m_i}$ that are contiguous in space as well as similar in intensity. This second partition breaks the spatial correlation of a purely intensity-based grouping: two voxels with the same intensity in different parts of the anatomy can land in different sub-regions and receive independent contrast perturbations. Each class is usually, but not always, sub-split this way (Sec. 7).

3. Signed affine remap. For each sub-region R , let $\bar{x}_R$ denote its mean intensity, $\mu_R \sim U(0, 1)$ a resampled target mean, and draw $\alpha_R \sim \pm U(0.5, 2.0)$ (a magnitude in $[0.5, 2.0]$ with an independently sampled sign). Every voxel $v \in R$ is remapped as

$$y_v = \mu_R + \alpha_R (x_v - \bar{x}_R). \quad (1)$$

Because Eq. (1) is a per-region affine function of the *real* intensity x_v , it preserves the relative structure of the signal within each region: edges, gradients, and partial-volume transitions between adjacent voxels are scaled, not replaced. Only the region-to-region contrast relationship is randomized, and $\alpha_R < 0$ additionally allows local contrast inversions that no acquisition protocol produces on its own, widening the space of contrasts the network sees beyond what any single real sequence could supply. Background voxels are zeroed at this step (and remain zero through the rest of the pipeline). The result is occasionally Gaussian-blurred, mostly not (Sec. 7).

4. Anatomical-label affine remap (optional). Steps 1–3 are unsupervised, so the k -means/Voronoi partition need not respect a true anatomical boundary: when two adjacent tissues share similar intensity, step 3 can by chance assign them the same target contrast and erase that boundary rather than randomizing it. This step corrects for that: for each real anatomical label ℓ present, we resample a fresh (μ_ℓ, α_ℓ) and apply Eq. (1) *again*, restricted to ℓ ’s voxels, **on top of the step-3 output** rather than the original voxels (Fig. 1, panel (d)) – so it composes with, rather than replaces, the unsupervised remap, and ℓ ’s voxels still inherit the internal structure steps 1–3 imposed on them. Because label assignment is independent of the unsupervised partition, this reintroduces a full-strength contrast discontinuity at the true label border regardless of whether step 3 preserved it, sharpening the boundary cue available to the network – not the underlying anatomy. It consumes only the labels the segmentation task already requires, so it needs no additional annotation, and is optional: disabling it (`label_remap_prob=0`) degrades PALETTE gracefully to steps 1–3 alone, e.g. on an unlabelled dataset. A final foreground z-score normalisation follows an optional second blur pass, giving the training input shown in panel (e) of Fig. 1. Per-step probabilities are set once, never tuned per dataset, and listed in full in Sec. 7.

3.2. Training protocol

PALETTE is applied on-the-fly as one contrast-augmentation operator inside an existing GPU-accelerated nnU-Net augmentation pipeline [22, 34], alongside that pipeline’s standard spatial and artifact augmentations; it adds no learned parameters and no measurable training-time overhead. A single configuration – the same k -range, sub-region count, blur schedule, and α range given in

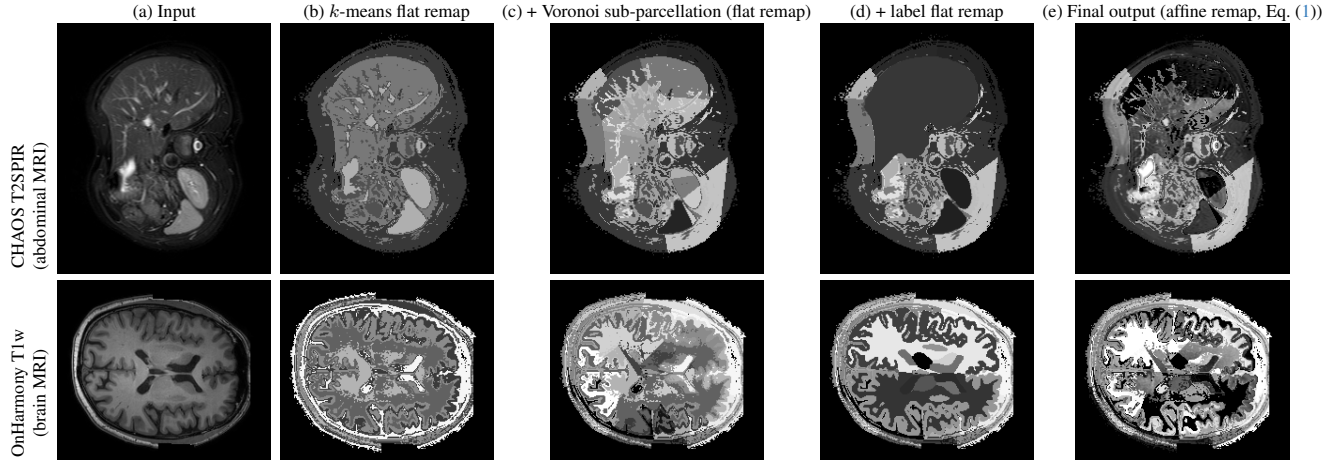


Figure 1. The PALETTE pipeline on one real CHAOS T2SPIR volume (top) and one real OnHarmony T1w volume (bottom); every panel is a deterministic function of the same real volume (no external model), min-max normalised to $[0, 1]$ for display only. Panels (b)–(d) paint each region, sub-region, or label with its precomputed target mean μ from Eq. (1), isolating that flat value from the texture applied on top of it; panel (e) is the actual training input, produced by applying the full signed affine remap of Eq. (1) with real texture (not a flat fill) at both the region and, where available, anatomical-label level.

§3.1 – is tuned once and reused unchanged across every dataset and modality in Sec. 4; no dataset receives its own hyperparameter sweep.

3.3. Causal ablation: isolating texture preservation

PALETTE differs from label-generative synthesis (e.g. SynthSeg) along two axes at once: the partition it perturbs (unsupervised k -means + Voronoi vs. a supplied anatomical label map as the sole generative substrate) and the perturbation itself (an affine remap of real intensities vs. resampling from a per-label Gaussian mixture). Comparing PALETTE to SynthSeg directly therefore cannot attribute an advantage to either axis in isolation.

We construct a controlled variant, *noise-fill*, that holds the k -means/Voronoi partition and the per-region statistics (μ_R, α_R) from Eq. (1) exactly fixed, but fills each sub-region with samples drawn from a Gaussian with that region’s $(\mu_R, \alpha_R \sigma_R)$ instead of remapping the real intensities inside it – i.e. the same partitioning scheme, with SynthSeg-style noise substituted for the affine remap. Because the partition and target statistics are identical to PALETTE’s own, any difference in downstream performance between PALETTE and noise-fill isolates the one variable that changed: whether real texture is preserved or discarded within each region. We report this comparison separately for structures with no tissue interface (fine internal gradients, low boundary contrast) and structures bounded by one (identified by a sharp, high-contrast border) in Sec. 4, as the texture-preservation hypothesis predicts a widening gap specifically on the former.

4. Experiments

4.1. Setup

Protocol. We train a single real-intensity source modality per dataset and evaluate on held-out modalities of the *same* segmentation task, using one fixed nnU-Net [22] configuration across all settings. No target-domain data, of any modality, is seen during training. Our full method, denoted **Ours** throughout, is PALETTE applied as one additional contrast-augmentation operator inside the Auglab pipeline [34]; the *Auglab* baseline is the same pipeline with the PALETTE operator removed, so the two differ by exactly the PALETTE transform. Every method is trained with the identical nnU-Net schedule — SGD with Nesterov momentum (0.99), the nnU-Net `polyLR` decay from an initial rate of 10^{-2} , 250 mini-batches per epoch, and the combined Dice-plus-cross-entropy loss — and differs only in the augmentation applied at the input. The epoch budget is fixed per dataset and shared by every method on that dataset (2500 for Brats-GLI, 2000 for ON-Harmony and Open-MS, 1000 for ToothFairy2 and I-SPY2, 200 for CHAOS and Pelvis; the Spine models were trained by our collaborators on their own pipeline with the same method suite, Sec. 8), and all results are 3-fold cross-validated, as is standard practice in medical image segmentation and so that every comparison rests on per-case statistics rather than a single split. All significance tests are paired per case and Holm-corrected [20]; full test construction, configuration details, and per-dataset descriptions (subject counts, acquisition protocols, and label provenance) are given in Secs. 7 to 9.

Task	Dataset	Train modalities	Test modalities
Multi-Organ	CHAOS	T1-in, T2-SPIR	T1-in, T1-out, T2-SPIR, CT
	AMOS	—	CT, MRI
	SLIVER07	—	CT
Brain Glioma	Brats2024-GLI	T1n, T1c, T2w, FLAIR	T1n, T1c, T2w, FLAIR
Brain MS	Open-MS	FLAIR, T1w	FLAIR, T1w, T2w
Healthy Brain	ON-Harmony	T1w, T2w, DWI	T1w, T2w, BOLD, DWI, EPI, GRE
Mandible	ToothFairy2	CBCT	CBCT
	HaN-Seg	—	CT, MR-T1
	PDDCA	—	CT
Breast	I-SPY2	T1-CE, T2w	T1-CE, T2w
	Duke-Breast-MRI	—	T1-CE, pre-contrast
	I-SPY1	—	T1-CE, pre-contrast
	ACRIN-6698	—	DWI
Spine [§]	CT/Dixon cohort	CT, Dixon in	CT, Dixon fat/in/water
	SPIDER	—	T1w, T2w
Pelvis	TotalSegmentator	CT, MRI	CT, MRI

Table 1. Per-task dataset roster. Rows marked “—” are cross-dataset transfer targets only, never trained on; their contrasts are extra held-out test columns of the task they sit under, not tasks of their own. Mandible’s held-out axis is entirely cross-dataset, since ToothFairy2 acquires CBCT alone. Pelvis’s CT and MRI come from different patients. [§]Spine results were computed by our collaborators on their own pipeline, with the same method suite but without SRCSM and without HD95 (Sec. 8). Subject counts, acquisition protocols, and label provenance are in Sec. 8.

Datasets. We evaluate on eight tasks drawn from sixteen datasets (Tab. 1): abdominal organs, healthy brain, brain tumour, MS lesions, mandible, breast lesions, spine and pelvic musculoskeletal anatomy, across MRI, CT and dental CBCT. Every task trains on a single source modality at a time and is tested on every other contrast of that task *and* the training contrast itself — eighteen source-modality settings in total; no target modality is seen during training. Wherever an external cohort exists, its contrasts are pooled into that task as further held-out test columns rather than reported as a task of their own: CHAOS [25] transfers to CT from AMOS [23] and SLIVER07 [19]; Mandible (ToothFairy2 [5], CBCT only) to the CT and MR of HaN-Seg [39] and the CT of PDDCA [41], which makes it the one task whose held-out axis is entirely cross-dataset; Breast (I-SPY2 [32]) to Duke-Breast-MRI [13, 42], I-SPY1 [21] and ACRIN-6698 [38], the last contributing diffusion-weighted MRI, a contrast neither training modality resembles; and Spine (a CT/Dixon cohort) to the T1w and T2w of SPIDER [51]. Pelvis is built from the TotalSegmentator CT and MRI releases [9, 54], two unpaired cohorts. ON-Harmony [53] reference labels are silver-standard, obtained on T1w with FreeSurfer [12] and registered to the other contrasts.

Baselines. SynthSeg comes in two variants, which we refer to throughout by how they obtain their per-class intensity model: *noEM*, the released variant, which reads it off a dense anatomical label map; and *EM*, which estimates it

from the image by expectation-maximization and therefore needs no dense parcellation. We compare against five methods. (1) *No-augmentation*, the nnU-Net default training recipe. (2) *SynthSeg-noEM* [3], the original label-generative variant (the implementation released by the authors), which synthesises each training image from a dense anatomical label map alone, ignoring the acquired scan entirely. (3) *SynthSeg-EM*, the variant described in the original work but never released publicly, which we re-implement; unlike SynthSeg-noEM it does not require a dense anatomical label map, but it is not dense-label-free either — it is compatible with and relies on the sparse task labels naturally available in a segmentation dataset, which it uses to estimate its per-class intensity model. (4) *SRCSM*, a semantic random-convolution contrast-synthesis augmentation, included as a second strong synthesis-based competitor that is neither label-generative nor a member of the Auglab family. (5) *Auglab*, the published GPU-accelerated augmentation pipeline of Molinier et al. [34]. Auglab is a prior method in its own right, not an ablation of ours, and we report it among the competitors accordingly; our full method (Sec. 4.1) applies PALETTE *on top of* the Auglab pipeline, so the PALETTE-vs-Auglab comparison also functions as the add-one-component test isolating the contribution of the PALETTE transform.

4.2. Main comparison

PALETTE is the best method overall. It attains the best Dice on six of the eight tasks, and across tasks it is significantly better than every competitor, including the strongest, Auglab ($p = 1.0 \times 10^{-13}$). Within individual tasks it significantly outperforms *every* competitor on four (Brats-GLI, Open-MS, Breast and Spine) and most of them on the remaining four (bottom row of Tab. 2). Every synthetic-contrast method vastly outperforms the no-augmentation baseline, confirming that cross-contrast augmentation is indispensable under single-source evaluation. SynthSeg-noEM, which resamples intensities from a *dense* anatomical label map, is competitive only where such maps are meaningful and collapses on the sparse-label tumour and lesion tasks, where no such map exists — a property of the method’s input requirements, not an implementation artifact. The re-implemented, label-using SynthSeg-EM closes most of that gap.

No competitor is reliably second. The strongest competitor alternates between two methods with opposite failure modes: Auglab on Brats-GLI, Open-MS, Breast and Spine, and SRCSM on CHAOS, ON-Harmony, Mandible and Pelvis — yet SRCSM collapses on Breast (16.4 Dice). Choosing between them would require knowing the answer in advance. PALETTE is first or second on every task but one. We regard this breadth, rather than the size of any sin-

	Brats-GLI	CHAOS	ON-Harm.	Open-MS	Mandible	Breast	Spine	Pelvis	p vs ours
Baseline	23.4	29.3	35.0	21.0	59.2	35.0	48.5	29.8	9.7e-75
SynthSeg-noEM	8.8	59.6	64.8	4.3	53.6	16.5	50.4	15.0	1.8e-85
SynthSeg-EM	43.1	83.0	66.3	33.9	76.3	39.6	80.6	62.5	1.4e-34
SRCSM	41.5	83.9	66.6	33.0	79.0	16.4	—	65.3	2.3e-86
Auglab	44.8	83.3	66.1	34.3	78.3	40.1	80.9	64.3	1.0e-13
PALETTE (ours)	45.9	85.0	66.5	38.5	77.7	41.2	82.3	65.6	—
competitors beaten ($p < 0.05$)	5/5	4/5	2/5	5/5	3/5	5/5	4/4	3/5	

Table 2. Task-level Dice (%). Each column pools that task’s held-out contrasts, all of its source modalities and, where one exists, its external cohorts, pooled by contrast group so that a modality tested by several external sources does not outvote one tested by fewer (Sec. 9). p : Holm-corrected one-sided sign-flip test on the macro-averaged per-patient difference, stratified by task with tasks weighted equally — a test *across* tasks, not an average *of* them; we deliberately print no cross-task mean Dice, since averaging a whole-liver Dice with a breast-lesion Dice yields a number no task’s clinician would recognise. The bottom row counts, per task, the competitors PALETTE beats with the same test run within that task alone. Spine has no SRCSM run (Tab. 1). HD95 is in Tab. 8.

gle margin, as the benchmark claim: a plug-in augmentation is useful precisely when it can be applied without knowing in advance which alternative would have been the right one for the task at hand.

Where PALETTE does not win, and by how much. The one-sided “ours is better” test cannot adjudicate a lead in the other direction, so on the two tasks PALETTE does not top we also ran the mirrored test, on the identical patient-level design. On ON-Harmony the result is a tie: SRCSM leads by 0.1 Dice and neither method is significantly ahead of the other ($p = 1.00$ both ways). On Mandible the loss is real: SRCSM (+1.3 Dice, $p = 6 \times 10^{-4}$) and Auglab (+0.6, $p = 2.5 \times 10^{-3}$) are both significantly ahead and PALETTE places third. The ablation of Sec. 4.3 locates that shortfall: on Mandible the texture step itself is significantly *positive*, and it is the subsequent addition of the Auglab recipe that costs Dice (Tab. 10). A benchmark ranking and a causal account are separate claims, and the second is the one this paper argues for.

4.3. Causal ablation: when does texture matter?

Boundary distance (secondary metric). HD95 is less standard than Dice for this class of comparison, so its task-level table is in the supplement (Tab. 8). In brief, PALETTE has the lowest HD95 on four of the seven tasks that report it (Brats-GLI, ON-Harmony, Open-MS, Pelvis) and is significantly better than every competitor across tasks, Auglab included ($p = 5.1 \times 10^{-6}$). It ties Auglab on Breast ($p \geq 0.43$ both ways), trails SRCSM on CHAOS without the gap being significant, and on Mandible is significantly behind SRCSM, Auglab and the no-augmentation baseline — the same task, and the same cause, as its Dice shortfall.

PALETTE shares its k -means/Voronoi partitioning scheme with a much simpler procedure, so the central question is whether the partitioning alone — not the texture-

preserving intensity remap — is what helps. We answer it with an intervention. Along a rung sequence in which each step adds exactly one ingredient (baseline $\rightarrow +k$ -means $\rightarrow +$ label-remap $\rightarrow +$ Voronoi with *noise* fill $\rightarrow$ same partition with *real*-intensity fill $\rightarrow +$ Auglab), every rung executes the identical pipeline of Sec. 3 in the identical order — the ladder only toggles which steps are active (e.g. the label-remap probability and the Voronoi sub-parcellation probability go from 0 to their default value one at a time), never reorders them. The fourth-to-fifth step changes *only the fill*: the partition, the number of sub-regions and the per-region statistics are held fixed, and label-style Gaussian noise is swapped for the real-intensity remap. The two Gaussian-blur passes (Sec. 3) are drawn from the same σ distribution at the same two points in the pipeline on both sides of this step, so blur is not confounded with the fill swap either. Any change at that step is therefore attributable to texture preservation alone.

Fig. 2 and Tab. 3 support a one-directional result. Every large gain from preserving real texture is on a target defined by tissue appearance rather than by an anatomical interface: Brats-GLI T1c (+13.3) and T1n (+7.2), both significant after correction over all fourteen rows, and Open-MS FLAIR (+7.7), positive for every one of its eight test patients. No tissue-interface row gains more than 1.4 Dice; the two that gain significantly, CHAOS T1in and ToothFairy2, do so by about a point. The converse does not hold, and we do not claim it: several appearance-defined settings show no reliable effect — Open-MS T1w, Brats-GLI T2w and FLAIR, and both Breast rows, where pooling five cohorts by contrast leaves the step at -0.85 and $+0.50$. A large texture gain is therefore diagnostic of an appearance-defined target, but an appearance-defined target does not guarantee one. This identifies *when* texture preservation can matter, and does so by intervention rather than by correlation across methods.

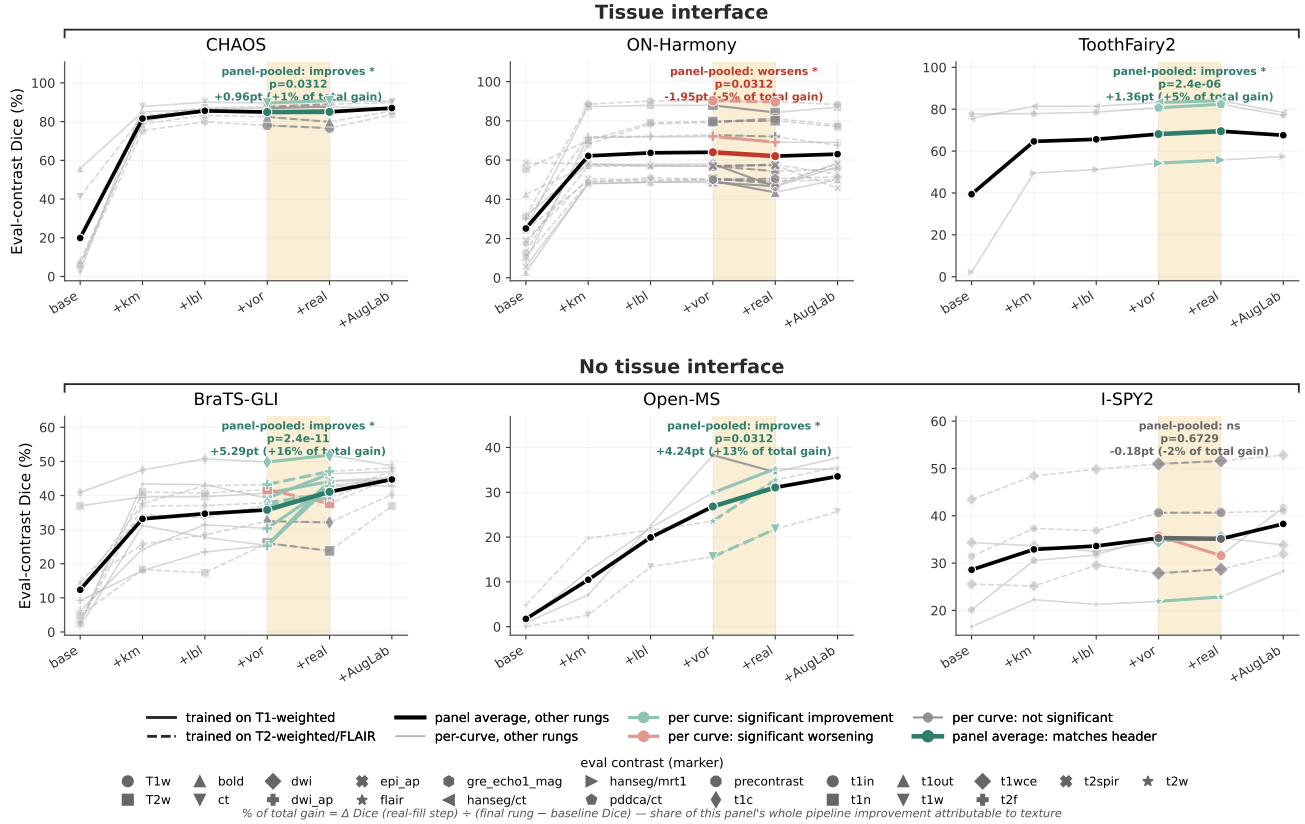


Figure 2. The one-variable test, one panel per task with a ladder, grouped by boundary type. Each thin line is one (training modality, held-out contrast) pair, external cohorts included, and only its real-fill segment (shaded band) is coloured, by that pair’s own significance at this one step – teal = significant improvement, red = significant worsening, grey = not significant. The bold black line is the panel average; its real-fill segment is coloured to match the panel-level test annotated above it, which pools all of the panel’s training modalities into one paired Wilcoxon test on *patients* (each counted once across contrasts and models), Holm-corrected across the six panels. Two of the three no-tissue-interface panels improve significantly (Brats-GLI $p = 2.4 \times 10^{-11}$, Open-MS $p = 0.031$) and Breast does not move. Among the tissue-interface panels, CHAOS and ToothFairy2 show small significant gains and ON-Harmony a significant worsening. Solid/dashed = trained on a T1-weighted vs. other contrast; marker = held-out contrast. Per-row numbers in Tab. 3; HD95 in Sec. 9.

Does ON-Harmony’s negative effect reflect the same tissue-interface pattern as CHAOS, or a different phenomenon driven by how densely it labels the brain? ON-Harmony’s 31 regions – the standard subcortical parcellation (lateral, 3rd and 4th ventricles, cortex, white matter, cerebellar white matter and cortex, brainstem, eight deep grey nuclei) – are, without exception, bounded by a genuine tissue-type transition: CSF against tissue for the ventricles, grey against white matter for cortex and nuclei. There is no texture-only label in the set, unlike a tumour or lesion margin, so ON-Harmony groups with CHAOS as a tissue-interface task, one that labels near-total brain volume rather than a handful of organs. Whether that labelling *density* has its own effect is a question one densely-labelled task cannot answer.

Does the fill swap change texture? Noise versus real fill.

The ladder shows that the fill swap changes Dice; it does not by itself show that what changed is texture. We measure that directly, without reference to any segmentation, using the normalized-gradient-field (NGF) similarity [16] between each augmented volume and its source scan — an alignment-of-gradients measure that is high when local edge and texture structure is retained and falls towards the isotropic-noise floor ($\approx 1/3$) when it is destroyed. Fig. 3 follows both quantities along the same rungs on Open-MS FLAIR. The three noise-fill rungs sit flat at the texture floor (NGF 0.45–0.46) while their Dice still climbs from 11.2 to 19.6: those gains come from richer contrast randomization, not from texture. Only the real-fill swap moves both together, NGF to 0.81 and Dice to 27.3. Table 4 places the methods on the same scale: PALETTE retains significantly more source texture than Auglab (FLAIR $p = 3.2 \times 10^{-3}$,

Task	boundary type	Δ Dice	p
CHAOS T1in	interface	+1.07	0.0014
CHAOS T2spir	interface	-0.88	0.17
ON-Harmony T1w	interface [†]	-5.18	0.078*
ON-Harmony T2w	interface [†]	-1.55	0.078*
ON-Harmony DWI	interface [†]	+0.78	0.11*
ToothFairy2 CBCT	interface [‡]	+1.36	5.8e-6
Open-MS FLAIR	no interface	+7.69	0.078*
Open-MS T1w	no interface	+0.79	1.0*
Brats-GLI T1n	no interface	+7.22	3.3e-11
Brats-GLI T1c	no interface	+13.27	5.0e-12
Brats-GLI T2w	no interface	+0.87	0.38
Brats-GLI FLAIR	no interface	-0.19	1.0
I-SPY2 T1-CE	no interface	-0.85	0.26
I-SPY2 T2w	no interface	+0.50	0.11

Table 3. Replacing label-style noise fill with the real-intensity remap, *holding the partition and per-region statistics fixed*: change in out-of-domain Dice for every training modality of every task with a ladder. p : paired Wilcoxon on patients, pooled over that row’s held-out contrasts, Holm-corrected across all fourteen rows. Every large gain is on a no-tissue-interface target; no interface row gains more than 1.4 Dice. *Open-MS has eight test patients and ON-Harmony eight test sessions (four subjects, each scanned at two sites), so no row of theirs can reach $p < 0.05$ after correction even when all eight move the same way, as they do for Open-MS FLAIR (8/8 improve) and ON-Harmony T1w and T2w (8/8 worsen). For ON-Harmony we treat the session as the unit; two sessions of one subject are not fully independent, so its p -values are, if anything, optimistic. [†]31 regions, each bounded by a genuine tissue-type transition (see text). [‡]Held-out axis is cross-dataset (HaN-Seg, PDDCA), since ToothFairy2 is CBCT only. HD95 in Sec. 9.

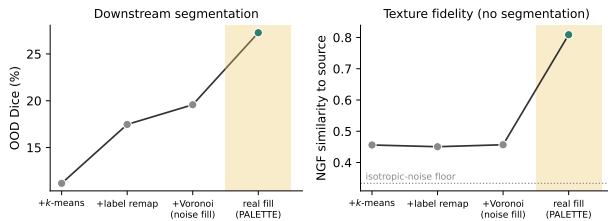


Figure 3. Noise versus real fill on Open-MS FLAIR. Left: out-of-domain Dice at each rung. Right: NGF similarity of the augmented volume to its source (mean $\pm$ s.e. over 30 scans), a segmentation-free measure of retained texture. The noise-fill rungs stay at the texture floor while Dice rises; the real-fill swap (shaded) lifts both.

T1w $p = 1.2 \times 10^{-6}$), while the label-style methods sit near the floor.

Why the margin over Auglab is the smallest. The same measurement explains why Auglab is the closest competi-

Method	FLAIR	T1w
PALETTE	0.809	0.894
Auglab	0.768	0.777
SynthSeg-EM	0.462	0.422
noise fill	0.457	0.401
SynthSeg-noEM	0.330	0.319

Table 4. Texture fidelity: NGF similarity to the source scan (mean over $n=30$ Open-MS scans; $\approx 1/3$ = isotropic-noise floor). PALETTE retains the most texture; the label-style methods sit near the floor.

tor: it is *not* a texture-destroying method, retaining NGF only modestly below PALETTE’s and far above the label-style methods’ (Tab. 4). Two methods that both largely preserve texture should land close together where texture is the operative cue, and they do; the large margins appear against the methods that discard texture.

Two properties of the test. First, it compares two versions of the same method that differ in one component, so it cannot be explained by differences in augmentation strength, partition granularity or training schedule. Second, it is independent of whether PALETTE wins the task. Mandible makes that concrete: the texture step is significantly positive there (+1.36), yet the full recipe places third, because the following rung — adding the Auglab pipeline on top — costs Dice ($69.5 \rightarrow 67.6$, Tab. 10).

Supporting analyses. A component ablation separating the k -means partition, the Voronoi sub-parcellation and the signed remap is in Sec. 11. Per-setting tables for all eighteen source-modality settings, every rung of every ladder and the HD95 versions of Tab. 3 are in Sec. 9.

5. Discussion and Limitations

The causal ablation isolates a mechanism, not a full explanation. The noise-fill comparison in Sec. 4.3 shows that texture preservation, specifically, drives the advantage on no-tissue-interface structures; it does not by itself explain the smaller, more uniform gains observed on tissue-interface structures, which likely stem from the wider space of region-to-region contrast relationships (including the sign inversions in Eq. (1)) that no single real acquisition protocol supplies. Disentangling these two effects further is left to future work.

Why the effect follows the anatomy, in one direction. Large gains from preserving real intensity structure appear only where the target is defined by tissue *appearance* rather than by an anatomical *interface*. Liver, spleen and kidney

are encapsulated organs bounded by fat planes – a genuine physical interface, which is why edge-following methods such as geodesic active contours have long succeeded on them [25, 47]; an augmentation that scrambles internal texture while preserving that interface loses little. Diffuse glioma has no such interface: tumour cells infiltrate beyond both the contrast-enhancing margin and the peritumoral FLAIR abnormality [17, 48], so the target must be inferred from appearance rather than followed along an edge. MS lesions are likewise defined by tissue signal characteristics rather than an anatomical border [57]. This account is an interpretation of the ablation, not an independent proof of it; the causal evidence is the intervention in Sec. 4.3. It also extends to the ON-Harmony result once its actual labels are examined (Sec. 4.3): its 31 regions – ventricles, cortex, white matter, deep grey nuclei – are bounded by genuine tissue-type transitions throughout, so despite first appearances it groups with the organ tasks rather than the no-interface ones. The larger negative magnitude there, compared to CHAOS’s near-null one, is plausibly a density effect – near-total brain volume relabelled at once, rather than a handful of organs – which we leave to a dedicated follow-up with a second densely-labelled task.

Appearance-defined is necessary, not sufficient. Several appearance-defined settings show no reliable texture effect: Open-MS T1w, Brats-GLI T2w and FLAIR, and both Breast rows. We do not have a tested explanation, and we state the result accordingly as one-directional. Two candidate factors are worth naming as hypotheses only. Within one task the effect differs sharply by training modality (Brats-GLI: +13.3 from T1c, +7.2 from T1n, about zero from T2w and FLAIR), so how much discriminative texture the *training* contrast carries may matter as much as the target. And Breast pools five cohorts from different trials and sites, which may dilute a cohort-specific effect. Either would be a property of the data rather than of the mechanism, but neither is established here.

The label step needs only the annotation training already assumes. PALETTE’s one label-consuming step uses whatever segmentation labels the training set happens to carry – sparse, partial, or task-specific – and never a dense anatomical parcellation. This is the practical difference from label-generative synthesis: SynthSeg’s dense requirement means the augmentation costs an annotation richer than the task, whereas here the step is free at training time and can be switched off entirely on an unlabelled dataset, in which case PALETTE degrades gracefully to its unsupervised steps.

Scope. PALETTE targets the single-source setting: one labelled training modality, no access to target-domain data

of any kind. It is input-level and architecture-agnostic, and we do not claim it subsumes multi-source domain generalization methods that have access to several labelled source domains; the two are complementary rather than competing solutions to the same problem.

6. Conclusion

We asked when preserving real image texture matters for contrast-robust segmentation, and found that the answer is a property of the task rather than of the augmentation: it matters where the target is defined by tissue appearance, and not where it is defined by an anatomical interface. A controlled intervention – holding the partition and per-region statistics fixed and swapping only label-style noise fill for a real-intensity remap – confines every large effect to no-tissue-interface targets — +13.3 and +7.2 Dice on brain tumour, +7.7 on MS lesions — while no interface-bounded target gains more than 1.4; it does not, conversely, guarantee a gain on every appearance-defined target. PALETTE, the single-source augmentation built on this observation, partitions the foreground by unsupervised k -means and a Voronoi sub-parcellation and applies a signed affine remap of the real intensities within each sub-region, requiring neither a dense anatomical label map nor any target-domain data. Across eighteen source-modality settings spanning eight tasks and sixteen datasets, a single nnU-Net configuration trained with PALETTE is the best method overall: best Dice on six of the eight tasks and significantly better than every competitor across tasks. It does not win everywhere: on healthy brain it ties SRCSM, and on mandible SRCSM and Auglab are genuinely ahead, a shortfall the ablation traces to the Auglab recipe PALETTE runs inside rather than to the texture-preserving transform. A benchmark ranking and a causal account are separate claims; we regard the task-level account as the more durable contribution: it predicts in advance where a texture-preserving augmentation will and will not pay off, and that prediction is testable on any new task.

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PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

Supplementary Material

7. Training configuration

All methods share one nnU-Net [22] configuration; only the input augmentation differs. Nothing below is tuned per dataset.

Training schedule. 3D full-resolution nnU-Net, SGD with Nesterov momentum 0.99, initial learning rate 10^{-2} with the nnU-Net polyLR decay, 250 mini-batches per epoch, combined Dice and cross-entropy loss, deep supervision on. Patch size, spacing and batch size are whatever the nnU-Net planner selects for that dataset, identically for every method. The epoch budget is fixed per dataset and shared by all methods on it: 2500 (Brats-GLI), 2000 (ON-Harmony), 2000 (Open-MS), 200 (CHAOS). Every number reported in this paper is a 3-fold cross-validated average; folds are the same across methods.

PALETTE hyper-parameters. Set once and never tuned per dataset: k -means class count drawn per sample from $\{2, \dots, 6\}$, skipped in favour of a single global remap with probability 0.10; Voronoi sub-parcellation of each intensity class with probability 0.60 (else left as one region); signed affine coefficient $\alpha \in \pm U(0.5, 2.0)$ applied per sub-region as in Eq. (1) (always applied, to whatever partition results from the k -means/Voronoi steps); per-label affine decoupling applied independently per label with probability 0.5 (skipped for any label with under 4 foreground voxels in that sample); Gaussian blur σ drawn from $\{0, 0, 0, 0.3, 0.5, 0.8\}$, applied twice (after the region-level and after the label-level remap), so mostly not applied at either point. The transform is applied on the GPU, on-the-fly, to a fixed 50% of training iterations (*train050*); the remaining half of iterations use the un-transformed real image.

Validation-time configuration. *val000* (the configuration reported in the main paper) applies no synthetic contrast to the validation split, matching every competing method and keeping model selection like-for-like. *val100* applies PALETTE to the entire validation split; it is reported in Sec. 10.

Baseline configurations. SynthSeg-noEM uses the authors' released generator with its default priors. SynthSeg-EM is our re-implementation of the variant described but not released in the original work; it estimates a per-class intensity model from the sparse task labels available in

each dataset via expectation-maximization, and is otherwise identical in its sampling procedure. SRCSM and Auglab use their published default configurations. No competing method was re-tuned in our favour, and all use the same folds, schedule and epoch budget as above.

Compute. Training was performed on NVIDIA L40S and H100 GPUs. A single fold costs from under an hour (CHAOS, 200 epochs) to roughly three days (Brats-GLI, 2500 epochs); the full study is dominated by the 7 methods $\times$ 8 settings $\times$ 3 folds main comparison plus the ablation ladders of Sec. 4.3.

8. Datasets and evaluation protocol

All evaluation is performed on held-out test cases only. For each task the model is trained on a single source contrast and evaluated on every contrast of that task's test split, including the training contrast itself; training and validation subjects never appear in any reported number.

ON-Harmony [53] — 31-region healthy-brain segmentation. Reference labels are *silver standard*: they were produced automatically with FreeSurfer [12] on the T1w volume of each subject and then registered to that subject's other contrasts, which are acquired in the same session. They are therefore not manual annotations, and absolute Dice on this task should be read as agreement with an automated reference rather than with a human rater; the comparison between methods, which all use the identical reference, is unaffected.

Open-MS [29] — MS-lesion segmentation with manual consensus annotations, evaluated across FLAIR, T1w and T2w.

Brats2024-GLI [10] — post-treatment glioma sub-region segmentation, evaluated across T1n, T1c, T2w and T2-FLAIR.

CHAOS [25] — abdominal organ segmentation (liver, spleen, both kidneys), evaluated across the CHAOS MR contrasts and, as a cross-modality transfer, on held-out CT from AMOS [23] and SLIVER07 [19].

ToothFairy2, HaN-Seg and PDDCA [5, 39, 41] — mandible segmentation, trained on dental cone-beam CT, the one non-CT, non-MRI training modality here. ToothFairy2’s annotation coverage is cohort-dependent, so every column of this task is scored on the mandible alone (teeth excluded, matching the external references), which keeps the numbers being averaged commensurable. ToothFairy2 acquires CBCT only, so its held-out evidence is entirely cross-dataset: HaN-Seg head-and-neck CT and MR-T1 and the PDDCA CT, the two CT sources pooled as one contrast group. ToothFairy2’s volumes are stored superior-inferior inverted relative to the external cohorts — an inversion invisible to the image header, found by checking anatomy directly — so the external test volumes are flipped into ToothFairy2’s convention before inference, which leaves Dice and HD95 unchanged under the rigid transform. The correction matters for the comparison: a method without flip augmentation (SRCSM) was penalised by the inversion far more than the others, and it is after this correction that SRCSM leads this task.

Breast: I-SPY2, Duke-Breast-MRI, I-SPY1, ACRIN-6698 [13, 21, 32, 38, 42] — breast-lesion segmentation on malignant-only cohorts. I-SPY2 (CC BY 4.0) trains, on its first-post-contrast T1 (T1-CE) and its T2w; its reference is the trial’s functional tumour volume mask. Three external cohorts are test-only. Duke-Breast-MRI and I-SPY1 come through the MAMA-MIA release, whose expert primary-lesion protocol both share, and contribute T1-CE and pre-contrast columns; Duke-Breast-MRI is CC BY-NC 4.0, a non-commercial term no other cohort here carries. ACRIN-6698, the diffusion sub-study of I-SPY2 with no patient in common with our I-SPY2 subjects, contributes the baseline diffusion-weighted acquisition with the trial’s whole-tumour DWI region, a contrast neither training modality resembles. Because I-SPY2 is mostly acquired unilaterally, bilateral external volumes are cropped to the lesion side and, for the full-thorax acquisitions, also anterior-posteriorly to the breast-plus-chest-wall extent of I-SPY2’s own crops; without the second crop every method’s HD95 is dominated by the thorax. I-SPY2’s DCE series are acquired unilaterally at some sites and bilaterally at others; both field-of-view variants are kept per patient, and the significance tests treat them as one patient.

Spine [51] — vertebrae, intervertebral discs and spinal canal. Results for this task were computed by our collaborators on their own pipeline with the same method suite, trained on CT and Dixon in-phase MRI of a CT/Dixon cohort and tested on that cohort’s held-out CT and Dixon fat, in-phase and water images and on the T1w and T2w of SPIDER. Their Dice is a whole-structure binary Dice per label; no HD95 and no SRCSM arm were computed, so Spine is

absent from the HD95 table and from every SRCSM comparison.

Pelvis [9, 54] — ten pelvic and hip musculoskeletal structures (both hips, sacrum, and the gluteus maximus, medius and minimus and iliopsoas on each side), from the TotalSegmentator CT (CC BY 4.0) and MRI (CC BY-NC-SA 2.0) releases. The two are unpaired cohorts, and the MRI release mixes sequences by design, so its MRI column is a sequence-heterogeneous test rather than one named contrast. CT is subsampled to 200 cases to keep training time comparable with the other tasks.

Task	Train	Test
CHAOS	16	24
ON-Harmony	84	8
Open-MS	22	8
Brats2024-GLI	630	70
ToothFairy2	408	71
I-SPY2	476	84
Pelvis (CT / MRI)	170 / 156	30 / 27

Table 5. Per-task subject counts for the training cohorts, pooled across the 3 folds used throughout (± 1 between folds). ON-Harmony counts are scan sessions; the eight test sessions are four subjects, each scanned at two sites. External cohorts are test-only and counted separately: AMOS and SLIVER07 (CHAOS), HaN-Seg and PDDCA (Mandible), Duke-Breast-MRI (291), I-SPY1 (167) and ACRIN-6698 (371) for Breast, and SPIDER (171 T1w, 219 T2w) for Spine, whose own CT/Dixon test split has 27–29 cases per contrast. I-SPY2 counts are patients; one held-out patient can yield more than one T2w image (both field-of-view variants). Pelvis MRI keeps 183 of the 298 released cases.

9. Per-setting results

Tab. 2 in the main paper reports results at the task level, pooling each task’s held-out test contrasts, all of its source modalities and its external cohorts. This section reports the same evaluation broken out into the eighteen individual source-modality settings. Every number is a cross-fold, cross-class average over all held-out contrasts of that setting, under the protocol of Sec. 4.1.

Statistical test construction. All tests are paired, and the unit of independence is the *patient*, never the fold, the image or the contrast. A patient’s score is its mean over labels and evaluated folds; where the same patient is scored in several held-out contrasts, or by the models of several training modalities, those scores move together as one unit under the null. Treating them as separate samples would be pseudo-replication and would inflate the effective sample size — a real risk here, since several tasks test a handful of patients

under many contrasts (ON-Harmony: eight sessions of four subjects, six contrasts). For ON-Harmony the unit is the session, since its four subjects leave a test that could never reach significance; two sessions of one subject are not fully independent, so its p -values are, if anything, optimistic, and we do not rest any claim on ON-Harmony alone. Per-setting comparisons (below) use the paired Wilcoxon signed-rank test [55], since there each setting is judged on its own. All tests are Holm-corrected [20] for multiple comparisons.

Why the aggregate tests are sign-flip tests on a macro-average. The aggregate tests (Tab. 2 and the per-dataset headline blocks) are not paired Wilcoxon signed-rank tests, which is the more usual choice in this literature: SynthSeg+, from the lineage we compare against, marks a method as better than its competitors using a one-sided Bonferroni-corrected Wilcoxon signed-rank test on paired Dice [4]. We therefore give our reason rather than leave it implicit. The *estimand* we test, by contrast, is not unusual at all — it is the same equal-weight-per-stratum principle the Medical Segmentation Decathlon adopted, for the same reason, as set out below.

The estimand. What we claim is $\text{macro}\Delta$: the mean, over held-out contrasts, of that contrast’s mean paired per-case difference (ours minus competitor) — equal weight per contrast, which is exactly what the *all* column of every table reports, so the test and the table measure the same thing. Equal weighting is not cosmetic here. The claim under test is cross-contrast generalization, and a contrast’s held-out case count reflects how large a public cohort happens to be, not how much that contrast should say about generalizing. CHAOS is the extreme case: its held-out CT evidence includes an external cohort contributing 200 cases, while each of its three MR contrasts contributes 8. Weighting by case count would let one CT cohort cast more than half the votes in a test whose conclusion is about transfer across MR contrasts.

This is the same problem, and the same answer, as the Medical Segmentation Decathlon’s. Facing sample sizes that “varied heavily across all tasks and target ROIs,” it ranked algorithms within each target ROI, averaged those ranks per task, and averaged the per-task ranks for the final score — choosing that over a flat average across ROIs precisely because the flat scheme means “tasks with multiple target ROIs ... would be over-weighted” [1]. Our $\text{macro}\Delta$ applies the same principle one level across, with the held-out contrast in place of the target ROI. Where we differ is only in how that weighted quantity is reached: the Decathlon arrives at it through counts of pairwise Wilcoxon outcomes at an uncorrected per-comparison α , whereas we test the weighted difference itself and Holm-correct across competitors.

Why not a pooled Wilcoxon. The signed-rank test is

defined on a single undifferentiated sample of paired differences. Applying it to the concatenation of all contrasts therefore tests a case-weighted null — the imbalance above, reintroduced — and there is no standard reweighting of the signed-rank statistic that recovers the equal-per-contrast estimand. The disagreement this causes runs in both directions: a pooled test can declare a tenth of a Dice point significant because one large cohort supplies most of the cases, or miss a consistent effect confined to the smaller contrasts.

The test. Under the null that the two methods are exchangeable on a given patient, that patient’s paired differences are equally likely to carry either sign. Sign-flipping is therefore a valid randomization null for *any* statistic computed from those differences [11], $\text{macro}\Delta$ included, which is what makes it applicable where the signed-rank test is not. We report the one-sided p in the direction of the stated hypothesis, with the two-sided p retained as a secondary column, and give a 95% confidence interval from a bootstrap that resamples patients within groups of patients sharing the same set of contrasts, so each source keeps its observed size.

Evaluating the tail. The statistic is a weighted sum of per-patient contributions, so its sign-flip null distribution can be computed rather than sampled. We enumerate all 2^M sign patterns of the $M=20$ patients with the largest contributions exactly, and treat the remaining, lighter patients as a Gaussian term with their exact null variance; with twenty patients or fewer the test is therefore exact. A plain normal approximation to the whole null, which an earlier version of this analysis used, is badly conservative in exactly the designs we have, where a few patients spanning several contrasts carry most of the null variance: on the CHAOS table it returned $p = 0.013$ for a comparison that a 200,000-draw Monte-Carlo test places below 5×10^{-6} . The enumerated tail is deterministic and reaches far tails without the $1/B$ floor of resampling. Its corollary is a hard floor for small cohorts: eight patients admit $2^8 = 256$ sign patterns, so no one-sided p can fall below $1/256$ however consistent the effect.

What is reported alongside. Every generated significance report also carries a per-contrast table of mean Δ with the raw, uncorrected Wilcoxon p for each contrast individually, and the per-setting tables in this section are Wilcoxon throughout. The standard test is reported; it is simply not the one the pooled headline number rests on.

Contrast-group pooling. For CHAOS, the same tested modality is sometimes evidenced by more than one source: CHAOS’s CT evidence comes from its own held-out CT split plus AMOS and SLIVER07. Averaging every raw column with equal weight — the natural first implementation — silently lets a modality with more contributing sources outvote one with fewer in both the displayed *all* average and the significance test, since it then supplies more of

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
Brats-GLI T1n	25.3*	8.7*	43.1*	40.3*	46.3	46.4
Brats-GLI T1c	28.3*	9.5*	42.9*	44.5*	46.6	47.1
Brats-GLI T2w	18.1*	8.4*	44.1*	43.8*	47.1	47.3
Brats-GLI FLAIR	21.8*	8.7*	42.3	37.5*	39.4*	43.0
CHAOS T1in	29.4*	62.6*	83.4*	83.7	83.1*	85.1
CHAOS T2spir	29.1*	56.6*	82.7*	84.1*	83.5*	84.8
ON-Harmony T1w	27.5*	66.0*	67.3*	66.4	67.6*	68.6
ON-Harmony T2w	34.8*	66.7	67.5	66.7	67.2	67.7
ON-Harmony DWI	42.6*	61.6	64.2	66.8	63.4	63.1
Open-MS FLAIR	25.2*	4.6*	32.8*	34.6*	32.8*	39.5
Open-MS T1w	16.7*	4.0*	35.1*	31.4*	35.8	37.4
Mandible CBCT	59.2*	53.6*	76.3*	79.0	78.3	77.7
Breast T1-CE	34.1*	16.6*	40.4*	18.0*	41.1*	41.9
Breast T2w	35.9*	16.5*	38.9*	14.8*	39.0*	40.5
Spine CT	27.0*	49.3*	79.1*	—	77.5*	80.2
Spine Dixon in	69.9*	51.6*	82.2*	—	84.2	84.3
Pelvis CT	35.6*	17.1*	66.5*	71.4	68.5	69.2
Pelvis MRI	24.0*	12.8*	58.6*	59.2*	60.0	62.0

Table 6. Per-setting Dice (%), $\uparrow$, best per row in bold. Each row is one training modality of one task, scored on exactly the held-out contrasts and external cohorts of its task column in Tab. 2, so the rows of a task average to that column. *: PALETTE significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run and no HD95.

the equally-weighted columns being averaged. We instead pool disjoint sources of the *same* modality by concatenating their cases into one group before any averaging happens, then average modality groups with equal weight – for CHAOS this additionally means averaging equal-weight across liver/spleen/kidney *within* the pooled CT evidence, since AMOS’s own CT column already blends all three organs into one number before it reaches this step. The same pooling applies to Mandible (HaN-Seg and PDDCA both supply CT) and Breast (I-SPY2, Duke-Breast-MRI and I-SPY1 all supply T1-CE and pre-contrast); the remaining tasks have no contrast evidenced by more than one source.

Tables 6 and 7 break each task column of Tab. 2 into its training modalities. Where a margin is not significant in a single setting, the honest reading is that one setting cannot separate the methods, not that they are equivalent: several settings have few test units (eight patients on Open-MS, eight sessions on ON-Harmony), which is why the main claims rest on the task- and cross-task-level tests.

Boundary distance at task level. Tab. 8 gives the HD95 counterpart of Tab. 2. Across tasks PALETTE is significantly better than every competitor, Auglab included. Per task it has the lowest HD95 on Brats-GLI, ON-Harmony, Open-MS and Pelvis; it ties Auglab on Breast and trails SRCSM on CHAOS without either gap being significant; and on Mandible it is significantly behind SRCSM, Auglab

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
Brats-GLI T1n	41.0*	78.0*	18.4*	20.8*	15.1	15.4
Brats-GLI T1c	40.3*	78.3*	19.5*	14.9	15.1	15.4
Brats-GLI T2w	54.9*	79.7*	16.5*	15.7*	14.8	15.0
Brats-GLI FLAIR	43.1*	76.0*	17.3	18.5*	19.2*	16.6
CHAOS T1in	120.0*	87.5*	33.8	26.3	43.5*	30.3
CHAOS T2spir	128.8*	92.4*	29.8	27.2	35.1*	30.9
ON-Harmony T1w	33.4*	10.4*	5.4	6.7	5.2*	4.3
ON-Harmony T2w	23.7*	4.6*	4.0	4.6*	4.2*	4.0
ON-Harmony DWI	14.7*	4.9	4.6	4.5	4.6	4.7
Open-MS FLAIR	39.9*	41.5*	23.6	23.8	26.6	24.3
Open-MS T1w	30.5*	42.3*	25.9	25.5	26.3*	24.2
Mandible CBCT	21.7	32.9*	24.9*	19.3	22.1	23.5
Breast T1-CE	61.8*	103.2*	61.7*	92.4*	57.3	59.0
Breast T2w	55.2	99.1*	60.1*	91.2*	56.9	56.3
Pelvis CT	146.2*	252.0*	20.0	16.0	23.5	21.6
Pelvis MRI	152.3*	271.6*	48.0	51.9*	47.8*	36.4

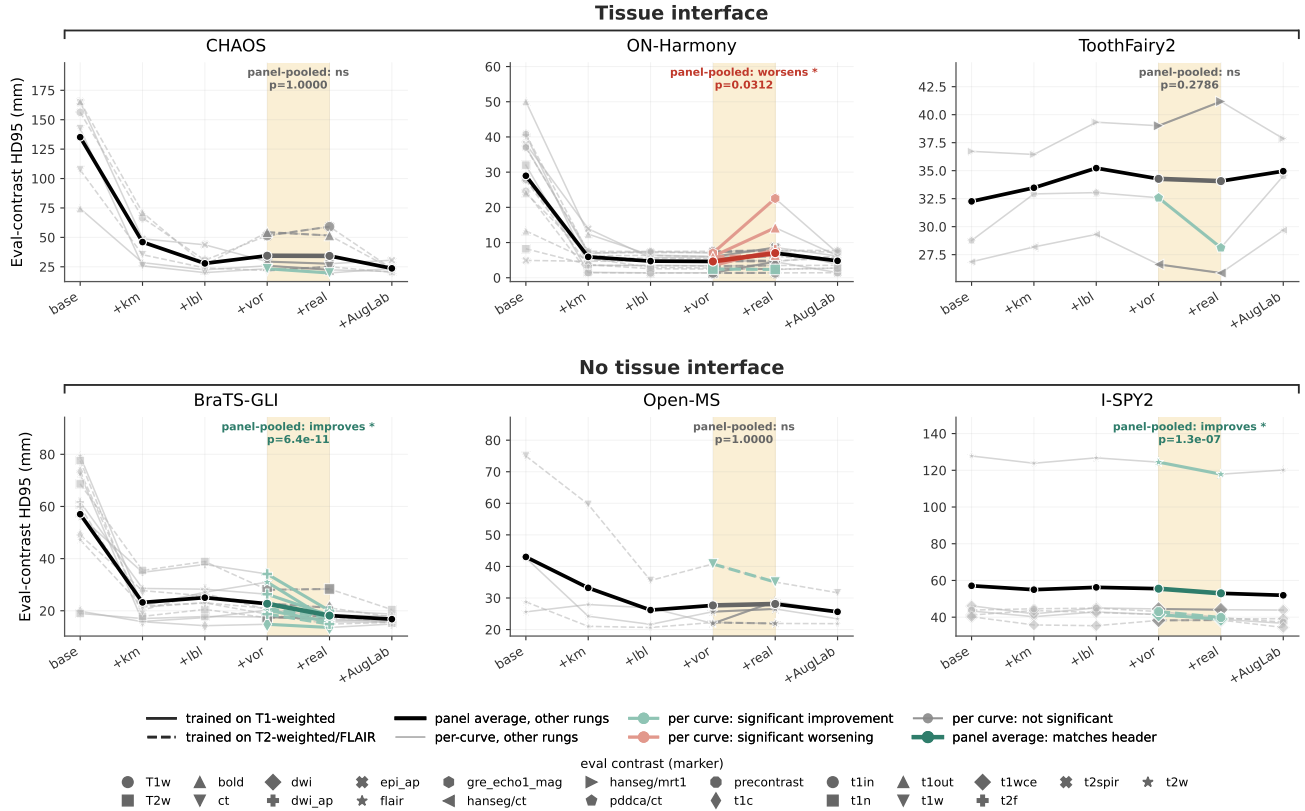
Table 7. Per-setting HD95 (mm, $\downarrow$), best per row in bold. Each row is one training modality of one task, scored on exactly the held-out contrasts and external cohorts of its task column in Tab. 2, so the rows of a task average to that column. *: PALETTE significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run and no HD95.

and the no-augmentation baseline, the one task where the baseline beats anything. When HD95 disagrees with Dice for a method – a noticeably better HD95 than its Dice would predict – the usual explanation is low recall: a method that predicts only a small, confident, well-localized fragment of the structure can have a deceptively good boundary distance while missing most of the target, so a strong HD95 paired with a weak Dice is a warning sign rather than evidence of competitiveness.

Full ladder: numbers and significance. Fig. 2 (main paper) shows the rung sequence visually; Tab. 10 gives the underlying out-of-domain Dice at every rung of all fourteen ladders, with significance for each step against the rung immediately before it (Holm-corrected paired Wilcoxon on patients, within each ladder’s family of steps). Most steps that add an ingredient help, as expected. Six cells are significant *decreases* (marked $\downarrow$). Two are the fill-swap step on ON-Harmony T1w and T2w, consistent with its tissue-interface grouping. One is the Voronoi sub-parcellation step on Brats-GLI T1n, which the real-fill step then more than recovers. The other three are the final +Auglab rung — on ON-Harmony DWI and twice on Mandible, the task where the complete method places third (Sec. 4.2). There PALETTE alone (the +real rung, 69.5) is the ladder’s maximum and adding the Auglab recipe takes it back down, which places the Mandible shortfall in the wrapper rather than in the transform this paper argues for. We have not diagnosed which Auglab operator is responsible.

	Brats-GLI	CHAOS	ON-Harm.	Open-MS	Mandible	Breast	Pelvis	p vs ours
Baseline	44.8	124.4	24.0	35.2	21.7	58.5	149.3	8.4e-33
SynthSeg-noEM	78.0	89.9	6.7	41.9	32.9	101.1	261.8	2.1e-39
SynthSeg-EM	17.9	31.8	4.7	24.7	24.9	60.9	34.0	4.7e-6
SRCSM	17.5	26.7	5.3	24.6	19.3	91.8	33.9	3.0e-23
Auglab	16.1	39.3	4.7	26.5	22.1	57.1	35.6	5.1e-6
PALETTE (ours)	15.6	30.6	4.3	24.3	23.5	57.7	29.0	—

Table 8. Task-level HD95 (mm, ↓). Same pooling and test as Tab. 2, and likewise no cross-task average. Spine reports no HD95.

Figure 4. The causal-ablation ladder of Fig. 2 (main paper), same per-dataset panels, same boundary-type grouping, same fill-swap rung and same visual scheme (per-curve colour only on the real-fill segment; bold black panel average; one panel-pooled p -value per panel), reported on HD95 instead of Dice (*improvement* means the fill-swap step lowers HD95). Brats-GLI and Breast improve significantly at panel level and ON-Harmony worsens; CHAOS, Open-MS and ToothFairy2 do not move significantly. Underlying per-row deltas in Tab. 9.

We likewise do not hide that Voronoi’s own marginal contribution is task-dependent rather than uniformly positive – a separate paired significance check (same protocol as Tab. 14’s all-contrast Dice, not just the OOD slice used above) confirms the same pattern: Voronoi significantly *helps* Open-MS FLAIR and (by a small margin) ON-Harmony T1w, is flat on CHAOS T1in, and significantly *hurts* Brats-GLI T1n under the real-fill deployed configuration as well, not only here under noise fill. The full method still wins on Brats-GLI T1n regardless (Tab. 2): whatever

Voronoi costs there is more than recovered by the subsequent real-intensity fill step, so the net effect of the complete pipeline is positive even though one internal component is not.

Per-label breakdown: is the fill-swap effect uniform within a task? Fig. 2 shows that Brats-GLI T2w’s pooled fill-swap effect is small and not significant, with one significantly negative eval contrast inside it (T2w→T1n). This raises a further question: within that one (training, eval)

Task	boundary type	Δ HD95 (mm)	p
CHAOS T1in	interface	-2.95	0.035
CHAOS T2spir	interface	+2.64	0.88
ON-Harmony T1w	interface	+6.82	0.062
ON-Harmony T2w	interface	+0.38	0.062
ON-Harmony DWI	interface	-0.05	0.75
ToothFairy2 CBCT	interface	-0.20	0.37
Open-MS FLAIR	no interface	-3.02	0.14
Open-MS T1w	no interface	+3.94	0.88
Brats-GLI T1n	no interface	-4.59	6.8e-10
Brats-GLI T1c	no interface	-9.91	4.5e-11
Brats-GLI T2w	no interface	-1.95	0.014
Brats-GLI FLAIR	no interface	-1.71	0.21
I-SPY2 T1-CE	no interface	-3.21	9.5e-5
I-SPY2 T2w	no interface	-1.82	0.0026

Table 9. Same fill-swap step as Tab. 3 (main paper), on HD95 (negative is better), with p Holm-corrected across the fourteen rows. Five of the six significant improvements are on no-tissue-interface targets, including both Breast rows, whose Dice does not move; the sixth is CHAOS T1in. HD95 therefore leans the same way as the main paper’s one-directional Dice result, but its sign is less consistent within a task (it flips between the two modalities of Open-MS and of CHAOS), so we state the claim on Dice and report HD95 here rather than as confirmation of it.

Rung	Open-MS FLAIR	Open-MS T1w	Brats-GLI T1n	Brats-GLI T1c	Brats-GLI T2w	Brats-GLI FLAIR	CHAOS T1in
base	2.4	1.1	16.5	15.9	6.0	11.0	21.4
+km	11.2*	9.8*	38.4***	29.6***	38.6***	26.1***	84.2***
+lbl	17.5*	22.4*	41.8***	30.3	40.2**	26.4	87.7***
+vor	19.6	34.1*	39.8***↓	30.5	41.0	31.9***	87.2
+real	27.3*	34.9	47.0***	43.7***	41.8	31.7	88.2***
+AL(v000)	30.6	36.4	46.9	44.2	46.8***	40.9***	87.7
+AL(v100)	31.3	34.7	47.6*	46.5***	46.9	41.2	87.3

Rung	CHAOS T2spir	ON-Harm. T1w	ON-Harm. T2w	ON-Harm. DWI	Mandible CBCT	Breast T1-CE	Breast T2w
base	18.3	14.8	24.5	36.1	39.4	23.7	33.5
+km	79.0***	62.5*	63.5*	60.4*	64.6***	28.8***	37.0***
+lbl	83.4***	62.9	63.6	64.5*	65.6***	28.5	38.8***
+vor	82.7	63.2	64.2*	64.6	68.1***	30.8***	39.8**
+real	81.8	58.0*↓	62.6*↓	65.4*	69.5***	29.9	40.3*
+AL(v000)	86.3	64.2	63.9	60.9*↓	67.6***↓	34.7***	41.9***
+AL(v100)	87.1	64.2	64.4*	60.9	67.5***↓	—	—

Table 10. Out-of-domain Dice (%) at every rung of every ladder (Fig. 2), with significance for each step against the rung above it: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (Holm-corrected paired Wilcoxon on patients, within each ladder); ↓ marks a significant *decrease*. Rungs: base(line), + k -means, +label-remap, +Voronoi (noise fill), +real-intensity fill (PALETTE alone), +Auglab (val000/val100). Open-MS and ON-Harmony have eight test units each, so their smallest attainable p after correction is just under 0.05. I-SPY2 has no val100 arm. Mandible’s held-out axis is cross-dataset.

pair, does the fill-swap step affect every anatomical sub-region equally, or is the worsening itself concentrated in specific labels? We break Brats-GLI T2w→T1n down by its four labels – necrotic core (NCR), oedema (SNFH), en-

hancing tumour (ET), resection cavity (RC); $n = 70$ test cases for SNFH and ET, present in essentially every case (NCR $n = 51$, RC $n = 67$, since not every tumour has a necrotic core or a resection cavity).

Label	T1n→T2w	T2w→T1n	gap
NCR	21.5	19.3	+2.1
SNFH	66.3	45.3	+21.0
ET	27.5	27.9	-0.4
RC	52.3	43.1	+9.1

Table 11. Absolute OOD Dice (%) of the deployed real-fill (v26.6.2) model, per label, both transfer directions.

Label	T1n→T2w Δ	T2w→T1n Δ
NCR	+1.9 (p=0.10)	+0.3 (p=0.87)
SNFH	+14.7 (p=4.3e-11)	-9.8 (p=4.5e-05)
ET	+2.3 (p=0.017)	+0.9 (p=0.38)
RC	+2.9 (p=0.0003)	-6.5 (p=0.0039)

Table 12. Fill-swap step (+voronoi noise fill → v26.6.2 real fill), per label, both directions. p : Holm-corrected paired Wilcoxon, family = the 4 labels within each direction. **Bold** = significant.

The worsening is not a property of “the tumour” – it is concentrated almost entirely in SNFH, which alone accounts for +21.0 of the roughly 14-point whole-tumour Dice gap between directions, and is the only label whose fill-swap delta significantly reverses sign between the two training directions (+14.7 when trained on T1n, -9.8 when trained on T2w). RC shows the same reversal at smaller magnitude. NCR shows no significant effect in either direction, and ET helps a little only in the direction that already works (T1n→T2w), never significantly reversing when trained on T2w. Fig. 5 shows why on one representative case: the T2w-trained model’s real-fill prediction on SNFH collapses to a thin band near the ventricle relative to its own noise-fill prediction on the same case, while its ET prediction looks essentially unchanged between the two rungs.

One candidate mechanism, flagged for future work rather than tested here: a texture pattern learned from one contrast’s rendering of a diffuse, texture-defined label like oedema may not simply fail to transfer to a different contrast’s rendering of the same tissue – it may actively mislead, if the model has come to rely on a contrast-specific cue that the other contrast does not reproduce. This would be an instance of shortcut learning [14] – specifically, a network relying on a domain-specific cue that does not reproduce across acquisition protocols, the mechanism Ouyang et al. [37] identify behind failed single-source domain generalization in cross-modality and cross-sequence medical segmentation – rather than the intended, causal feature, which

BraTSGLI02108101, fold 0 -- T2w-trained model evaluated on T1n

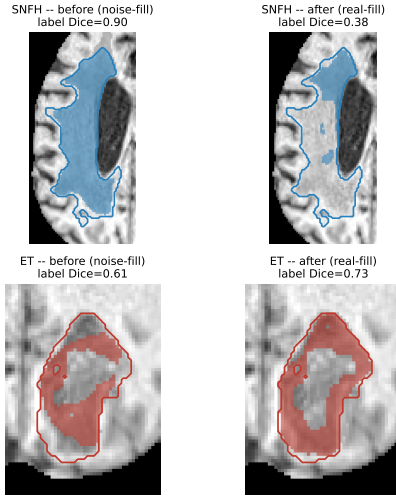


Figure 5. BraTSGLI02108101, fold 0, T2w-trained model evaluated on T1n. Zoomed crops before (noise-fill) and after (real-fill) the fill-swap step, for the SNFH-dominant (top) and ET-dominant (bottom) lesion components of the same case. Colour = ground truth wherever the prediction agrees; grey = false positive; outline = ground truth.

a simplicity-biased network need not prefer even when it would generalize better [44]. We have not tested this mechanistically here; it is the most concrete hypothesis consistent with the label-specific, direction-dependent pattern above, not a demonstrated cause.

10. Synthetic contrast at validation time

Throughout the main paper PALETTE is applied with synthetic contrast during training only (*train050/val000*), because every competing method is likewise evaluated with a real-contrast validation set; this keeps the comparison exactly like-for-like. PALETTE can additionally be applied to the validation set, so that model selection is performed under randomized contrast as well (*train050/val100*).

Validation-time randomization improves most tasks on both metrics (Tab. 13), most visibly on boundary distance for CHAOS (30.6 $\rightarrow$ 28.7 mm). The effect is not significant across tasks, and Open-MS Dice moves slightly the other way, so we do not claim it as a separate contribution; we report it because it costs nothing and because it indicates that the benefit of contrast randomization extends to model selection and is not confined to the gradient updates.

11. Component ablation

To attribute the gain of the full method to its individual components, rather than to synthetic-contrast augmentation in

Task	Dice $\uparrow$		HD95 (mm) $\downarrow$	
	val000	val100	val000	val100
Brats-GLI	45.9	46.6	15.6	15.1
CHAOS	85.0	85.2	30.6	28.7
ON-Harmony	66.5	66.6	4.3	4.3
Open-MS	38.5	37.9	24.3	24.2

Table 13. Applying PALETTE at validation time as well as at training time, on the four tasks where both configurations were evaluated (Mandible’s val100 arm was not evaluated on its cross-dataset test sets; the Breast, Spine and Pelvis arms were not trained). Across these tasks the two are statistically indistinguishable in either direction (same cross-task test as Tab. 2; two-sided Dice $p = 0.62$, HD95 $p = 0.18$), though val100 is significantly ahead within Brats-GLI and ON-Harmony individually. We report the like-for-like val000 configuration throughout the main paper.

Component	CHAOS	Open-MS	BraTS	ON-Harmony
Base (no augmentation)	36.1	25.2	25.3	27.5
+ k -means partition	84.8	30.6	40.6	67.6
+ label remap	87.7	34.6	43.2	68.0
+ Voronoi sub-parcellation	87.2	36.1	41.8	68.2
+ affine remap (Ours)	87.9	40.4	47.4	63.6
+ Auglab (deployed)	87.4	39.5	46.4	68.6

Table 14. Mechanism ablation (top five rows, additive, no Auglab) and deployment composability check (bottom row). Each rung adds exactly one PALETTE component on top of the row above; all-modality Dice (%), averaged over held-out test contrasts, folds 0–2, train fraction 50%. BraTS and ON-Harmony report their T1n/T1w settings, matching every other table in this paper. The deployed row matches Tab. 6’s **Ours** column exactly.

general, we build an additive ablation ladder on top of the no-augmentation anchor (*Base*, Sec. 4.2): first the unsupervised k -means intensity partition alone (§3.1, step 1); then the anatomical label-remap step (step 4); then the Voronoi spatial sub-parcellation (step 2); and finally the full signed affine remap of Eq. (1) (step 3), which together constitute PALETTE in its entirety and is denoted **Ours** in this ladder — identical to the standalone (no-Auglab) configuration used throughout Sec. 4.2. Every rung is evaluated without Auglab, so that the ladder isolates PALETTE’s own internal components from the separate question of whether it composes with a prior augmentation pipeline, which we address next. The synthetic-data fraction is fixed at 50% of training iterations throughout, as one of several augmentation constants (alongside, e.g., the k -means class count and the affine magnitude range of §3.1) that are set once and not tuned per dataset; validation uses real-contrast data only (no synthetic contrast at validation time).

1476 **Composability with Auglab.** PALETTE is deployed,
1477 throughout this paper, as one additional operator inside
1478 the Auglab pipeline rather than as a replacement for it
1479 (Sec. 4.1); Auglab is a full prior augmentation method in
1480 its own right, not a component of PALETTE, so stacking
1481 it on top of the ladder above is a composability/deployment
1482 check, not a further mechanism-ablation step. Tab. 14 keeps
1483 the two separate: the top five rows isolate PALETTE’s in-
1484 ternal mechanism, evaluated with no Auglab throughout,
1485 while the bottom row reports PALETTE plus Auglab — the
1486 configuration used everywhere else in this paper — set apart
1487 by a rule rather than presented as the next rung of the same
1488 ladder.